ENGINEERING
PROFILE / 01

CONTACT

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Portfolio / Bangladesh

TECHNICAL TOOLKIT

PROGRAMMING

Python, Dart, PHP
Embedded C/C++ fundamentals

AI / COMPUTER VISION

TensorFlow, Keras, CNNs
Transfer learning
Image classification
Model evaluation

HARDWARE

ESP32, Arduino UNO
Sensor integration
I²C, relay control
Embedded automation

APPLICATIONS

Flutter, FastAPI
REST APIs
TensorFlow Lite

WORKFLOW

Git/GitHub, VS Code
Google Colab
Anaconda/Conda
Linux fundamentals

MORE PROJECT SOURCE

FarmGuard-AI
ROS 2 / Gazebo
SMS classification

ROBOTICS / AI / EMBEDDED SYSTEMS

AL AMIN

IoT & Robotics Engineering Undergraduate

Building experience where intelligent software meets physical systems.

01 ENGINEERING FOCUS

Fourth-year undergraduate targeting robotics and AI engineering roles, with hands-on project experience in computer vision, machine learning, and microcontroller-based automation.

02 EDUCATION

B.Sc. in Internet of Things and Robotics Engineering (IRE)

University of Frontier Technology, Bangladesh (UFTB)
4th year, 1st term (4-1)

03 PROJECT EXPERIENCE

Neuro-Symbolic Rice Leaf Disease Detection

AI-based disease detection with mobile application integration.

- Audited and preprocessed datasets; trained and evaluated CNNs, comparing MobileNetV2, DenseNet121, and EfficientNetB0.
- Developed the disease detection and reasoning workflow; worked on Flutter, FastAPI, and TensorFlow Lite integration.

Status: Core model development and evaluation completed; mobile integration and documentation ongoing.

IoT-Based Smart Home Automation

ESP32 + Arduino UNO | Prototype developed

- Implemented sensor interfacing, microcontroller communication, relay control, and automatic control logic.
- Published an ESP32-UNO UART implementation with Adafruit IO commands, automatic sensor rules, and a browser telemetry dashboard.

04 RECOGNITION

Mathematics Olympiad: Dhaka North achievement (2024); 15th Olympiad final-round participation (April 2025); 16th Olympiad regional participation (November 2025).

ROBOFUSION 1.0: Participation certificate, UFTB Robotics Club.

TEDxUttara 2025: Attendance certificate (December 2025).

05 RESEARCH & TEACHING

Dataset analysis, experimentation, literature review, and technical documentation through undergraduate projects. Private tutoring experience.